TCS-662(MACHINE LEARNING)

ASSIGNMENT-4

Q1. Explain the K-means clustering algorithm with suitable example. Discuss advantages and limitations. CO4

Q2. Describe the Elbow Method and Silhouette Score for determining optimal number of clusters. CO4

Q3. Compare Agglomerative and Divisive Hierarchical Clustering with examples. CO4

Q4. Discuss the working of DBSCAN with an example. How does it detect outliers? CO4

Q5. Given covariance matrix: CO4

6	2
2	5

Find eigenvalues for PCA.

Q6. Explain Principal Component Analysis (PCA). CO4

Q7. Describe Linear Discriminant Analysis (LDA) and compare it with PCA. CO4

Q8. Explain Forward Selection and Backward Elimination methods of feature selection. CO4

Q9. In a given Table, Identify Core, Border, and Noise Points (DBSCAN) CO4

Point	Coordinates
A	(1,1)
B	(1,2)
C	(2,1)
D	(8,8)
E	(8,9)
F	(25,25)

Let: Eps = 1.5, MinPts = 3 Find: Core points, Border points, Noise points, Final clusters

Q10. Discuss the Curse of Dimensionality and its impact on clustering and classification. CO4

Q11. Explain the working of a Perceptron. What is the difference between single-layer and multilayer neural networks? CO5

Q12. Describe different activation functions used in neural networks. CO5

Q13. Explain Backpropagation Algorithm with examples. CO5

Q14. What is a Convolutional Neural Network (CNN)? Describe the working of convolution operation in image processing. CO5

Q15. Explain the architecture and applications of autoencoders. CO5

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